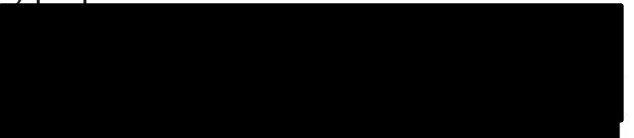


Department of the Army
Justification and Approval for Other Than Full and Open Competition

1. **Contracting Activity:** The United States Army Contracting Command-Detroit Arsenal (ACC-DTA) is the procuring activity for this action.

2. **Description of Action:** ACC-DTA proposes to award a new Firm Fixed Price Contract to Surface Optics Corporation, Commercial and Government Entity Code 1V827, to procure the SOC-410 Directional Hemispherical Reflectometer (OHR), Part Number 410-0032#C. Surface Optics Corporation is located at 11555 Rancho Bernardo Road, San Diego, CA, 92127-1441 and is a small business. The type of funds to be used for this action are Fiscal Year 2024 Research, Development, Test, and Evaluation.

3. **Description of the Supply or Service Purchased:** DHR's are commercially manufactured to measure the reflectance of various surfaces using a software known as Engineering mode. Engineering mode is capable of displaying reflectance data of Chemical Agent Resistant Coatings (CARC) properties in numerical form which is stored on a removable Secure Digital card 


Furthermore, the OHR tool will have an additional software package known as Quality Control mode. Quality Control mode will display unclassified feedback to non-technical operators while performing quality control checks with pass or fail output via the DHR tool computer display. The default mode of operation will be Quality Control mode with a password required to enter Engineering mode. Quality Control mode is needed to provide simple feedback for different end users to verify if the CARC paint system is functioning as expected. The purpose of Quality Control mode is to mitigate mishandling of reflectance data regarding the CARC paint system.

The Government intends to procure eight (8) handheld DHR tools with Engineering and Quality Control mode software packages to support the requirement for Product Manager (PdM) Vehicle Protection Systems (VPS). The total estimated cost is \$761,000.00, which is based on the Independent Government Cost Estimate (IGCE) developed by the program office on 22 August 2024.

4. **Authority Cited:** The statutory authority permitting other than full and open competition is 41 United States Code (U.S.C) 1901 as authorized and implemented by Federal Acquisition Regulation (FAR) 13.501(a) "Sole Source (including brand name) acquisitions".

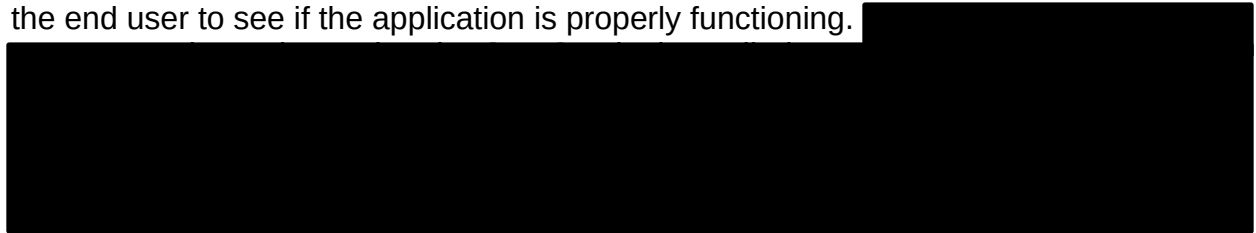
5. Reason for Authority Cited:

Surface Optics Corporation is the Original Equipment Manufacturer for the only fully portable hand-held device for quick inspection of coatings and materials in the field. Surface Optics Corporation developed the DHR tool in collaboration with U.S. Naval Air Systems Command with input from the Naval Research Lab and National Institute of Standards and Technology. DHR was developed with the capability for different software options to be developed since this item has absolute reflectance which can measure various materials, metals, electrics, or other surfaces including CARC paint systems.

Surface Optics originally developed the SOC-410 (Part Number 410-0032) under Small Business Innovation Research (SBIR) topic number: N01-141. The Phase III SBIR contract (N41756-04-D1356) was developed in conjugation with the U.S. Navy. Due to the hardware and software of the original model developed under this SBIR being “discontinued/end of life”, the original SOC-410 DHR (410-0032) for which the Government had the Technical Data Package (TDP) is now obsolete. The current model of the tool is now classified as Part Number 410-0032#C.

Surface Optics Corporation is the only source who produces the current DHR tool and owns the technical data along with both softwares mentioned. On 18 October 2024, the VPS Contracting team requested to enter negotiations with Surface Optics to acquire their Technical Data Package and associated software rights required to compete the DHR tool. Surface optics declined to enter this agreement as they will not allow any other party to manufacture the tool.

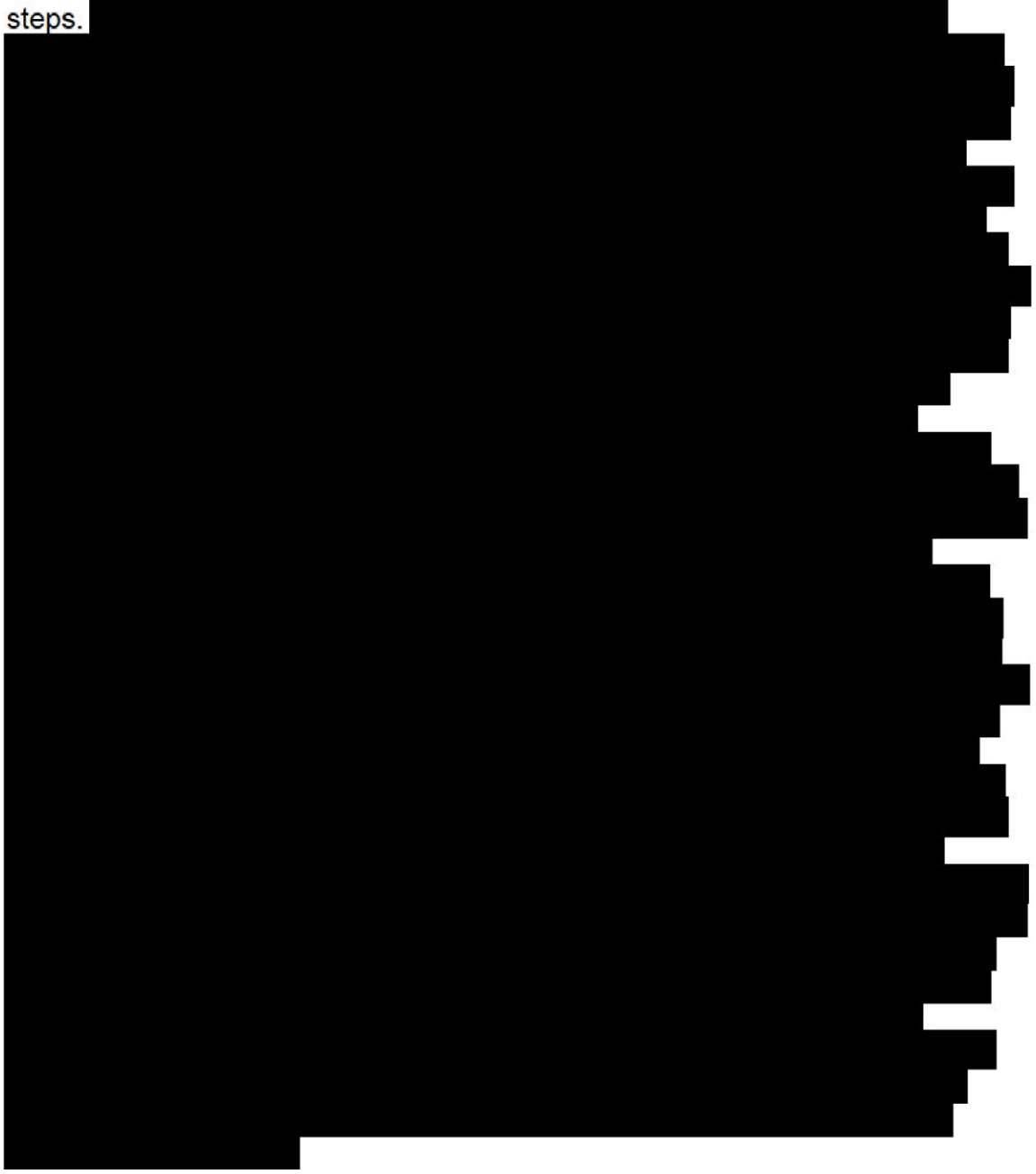
With new developments of CARC application efficiency, it is important there are no delays in award of this contract which could result in serious vulnerability or injury of the warfighter without proper quality checks. In addition, Signature Management (SIGMAN) Paint Program has high visibility with a priority of continuous efforts to improve efficiency of the CARC application process. The DHR tool has been identified as a need at the end of CARC application process for quality purposes. During the Quality Assurance phase of CARC application, the DHR tool will be instrumental and will allow the end user to see if the application is properly functioning.



The use of FAR 13.501(a) is justified for the following reasons:

- a. According to estimates by the VPS Lead Engineer, it would take the Government 3 years and 8 months to reverse engineer this tool. Estimates for the tool to be reversed engineered were created based on five (5) different

steps.



- b. In addition to the prolonged timeframe above, the cost estimate to reverse engineer the tool is \$2,479,050. If the Government was to compete this action with a TOP derived from the reverse engineered tool, we could expect to see a savings of \$5,000 per device bought. With the current price of the device estimated around \$95,000, the Government would spend \$2.5 Million to reverse engineer then \$720,000 to acquire the 8 OHR tools at a \$5,000 discount per unit. The Government would have to acquire 500 more OHR's to recover the costs from the reverse engineering. Currently, the Government has plans to potentially acquire a few more OHR tools over the next 5 years, but not nearly enough to recoup the costs of the reverse engineering.

[REDACTED]

	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]					[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]					[REDACTED]
[REDACTED]	[REDACTED]				[REDACTED]

6. Efforts to Obtain Competition:

- a. *Effective Competition.* The Government is not pursuing competition for this requirement based on the reasons cited in section 5 above. All notices required by FAR 5.201 will be posted and all proposals from responsible sources will be considered.
- b. *Subcontracting Competition.* Surface Optics Corporation is considered a small business, as such a subcontracting plan will not be required under this award.

7. **Actions to Increase Competition:** ACC-DTA will continuously survey the market and perform market research in the future to see if there are new responsible parties to produce the OHR. The Government will continue to post Requests for Information

(RFI's) to SAM.gov to survey whether another contractor is capable of producing the DHR Tool with Engineering mode and Quality Control mode software. At this time Surface Optics is the only company capable of producing the DHR tool with the add-on software the Government needs. We do not anticipate this will change in the near future.

8. Market Research: PdM VPS and ACC-DTA conducted market research to determine what capabilities and who in industry could fulfill this requirement. A RFI was posted on the SAM.gov website on 13 March 2024 for 30 days. While the RFI was open, the Government only received one response. Surface Optics Corporation was the only vendor who expressed interest and responded to the RFI with their capabilities. In the past, Surface Optics Corporation has received six prime contracts to deliver handheld DHR tools to different Government Agencies. In addition, Surface Optics Corporation confirmed through their RFI response they are able to develop and have successfully produced two unclassified modes to fulfill Governments needs of protecting data during quality checks.

SIGMAN Lead Engineer conducted market research after the closing of the RFI using World Wide Web searches to determine the availability of the DHR tool. During this research it was concluded that Surface Optics Corporation is the only source. SIGMAN Assistant Program Manager (APM) also conducted market research via usaspending.gov on 11 July 2024 for previous procurement information related to cost while developing the IGCE. It was concluded based on all market research conducted that Surface Optics Corporation was the only contractor able to produce the DHR tool, develop and incorporate software to meet the requirements specified by the SIGMAN Team/PdM VPS.

9. Interested Sources: An explanation of interested sources is detailed in section 8 above. To date, no other sources have written to express an interest

10. Other Facts:

a. Procurement History

- 1) Contract Number: W911QX-21-P-0116
Award Date: 25 May 2021
Contractor: Surface Optics Corporation
Price: \$91,423.85
- 2) Competitive Status: Sole Source (Confirmed via SAM.gov Award Notice)
- 3) Authority Cited: Not awarded by Army, information unavailable
- 4) Justification: Not awarded by Army, information unavailable
- 5) Full and Open competition has not been used on prior contracts for this Requirement
- 6) There has been no unusual patterns in previous buys of the SOC-410 DHR
- 7) Justification of previous contract not available as this is the first time the Army is procuring this tool

11. Technical Certification: I certify that the supporting data under my cognizance, which is included in the justification is accurate and complete to the best of my knowledge and belief.

Printed Name: Randolph Nasir	Electronic Signature:	OPSEC #17 PII
Title: CCDO Lead Engineer	Date: 8/14/2024	

12. Requirements Certification: I certify that the supporting data under my cognizance, which is included in the justification is accurate and complete to the best of my knowledge and belief.

Printed Name: Sarah Schlauch	Electronic Signature:	OPSEC #17 PII
Title: APM (Acting)	Date: 08/14/2024	

13. Fair and Reasonable Determination: I hereby determine that the anticipated cost to the Government for this contract action will be fair and reasonable. The negotiated price will be determined fair and reasonable through a combination of a Cost Analysis to thoroughly analyze each of the proposed individual cost elements in addition to conducting a Price Analysis to ensure that the price agreement is consistent with the Independent Government Cost Estimate and/or historical pricing for similar items.

Printed Name: Jennifer S	Electronic Signature:	OPSEC #17 PII
Title: Contracting Officer	Date: 11/26/2024	

14. Contracting Officer Certification: I certify that this justification is accurate and complete to the best of my knowledge and belief.

Printed Name: Jennifer Harris	Electronic Signature:	
Title: Contracting Officer	Date: 11/26/2024	

APPROVAL

Based on the foregoing justification, I hereby approve the procurement of eight (8) OHR tools with Engineering and Quality Control mode software packages for a total estimated value of \$761,000.00 on an other than full and open competition basis pursuant to 41 U.S.C 1901 as authorized and implemented by FAR 13.501(a) "Sole Source (including brand name) acquisitions". This procurement is subject to the availability of funds, and provided that the supplies or services herein described have otherwise been authorized for acquisition.

X

Rachel Capaldi
Command Advocate for Competition